

Electronic Workshop 2: Audio Amplifier Design Report

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1 Introduction

This report details the design and simulation of a multi-stage amplifier circuit. The system is designed to process an audio signal through four distinct stages: a Pre-amplifier, a Gain Stage, a Band Pass Filter (BPF), and a final Class AB Power Amplifier stage to drive a load.

2 Pre-Amplifier

2.1 Circuit Topology

The pre-amplifier stage is implemented using a **BJT Differential Amplifier**. This configuration was selected because it provides:

- Moderate and controllable voltage gain
- High input impedance
- Inherent common-mode noise rejection
- Improved bias stability compared to single-ended CE stages

The circuit consists of two matched NPN transistors sharing a common emitter resistance (R_E) connected to the negative supply rail. Equal collector resistors (R_C) are used, and the output is taken single-ended from one collector.

2.2 Large-Signal (DC Bias) Analysis

For DC analysis, all capacitors are treated as open-circuits.

The tail current is determined by the emitter resistor:

$$I_E = \frac{|V_{EE}|}{R_E} \quad (1)$$

With $V_{EE} = -5V$ and choosing:

$$I_E = 3 \text{ mA} \quad (2)$$

$$R_E = \frac{5}{3} = 1.67 \text{ k}\Omega \quad (3)$$

The nearest standard value selected:

$$R_E = 1.5 \text{ k}\Omega \quad (4)$$

Since the current splits equally:

$$I_C = \frac{I_E}{2} = 1.5 \text{ mA} \quad (5)$$

The collector DC voltage becomes:

$$V_C = V_{CC} - I_C R_C \quad (6)$$

The Q-point is selected near the midpoint of the load line to allow maximum undistorted swing.

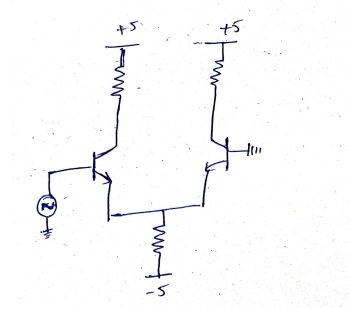


Figure 1: Large-Signal (DC Bias) equivalent circuit of the Differential Amplifier.

2.3 Small-Signal Gain Derivation

For small-signal analysis:

- DC sources are grounded.
- The hybrid- π model is used.

Let the differential input be:

$$v_1 = -v_2 \quad (7)$$

The emitter node voltage is denoted as v_e .

Step 1: KCL at the Emitter Node

Applying KCL at the emitter node:

$$\frac{v_{\pi 1}}{r_{\pi}} + \frac{v_{\pi 2}}{r_{\pi}} + g_m v_{\pi 1} + g_m v_{\pi 2} = \frac{v_e}{R_E} \quad (8)$$

Since:

$$v_{\pi 1} = v_1 - v_e \quad (9)$$

$$v_{\pi 2} = v_2 - v_e \quad (10)$$

For differential input:

$$v_2 = -v_1 \quad (11)$$

Thus:

$$v_{\pi 1} + v_{\pi 2} = (v_1 - v_e) + (-v_1 - v_e) = -2v_e \quad (12)$$

Substituting:

$$\left(\frac{1}{r_\pi} + g_m \right) (-2v_e) = \frac{v_e}{R_E} \quad (13)$$

$$v_e \left[2 \left(g_m + \frac{1}{r_\pi} \right) + \frac{1}{R_E} \right] = 0 \quad (14)$$

For sufficiently large R_E :

$$2 \left(g_m + \frac{1}{r_\pi} \right) \gg \frac{1}{R_E} \quad (15)$$

Hence,

$$v_e \approx 0 \quad (16)$$

Thus each transistor sees half of the differential input:

$$v_{\pi 1} = \frac{v_{id}}{2} \quad (17)$$

Step 2: Output Voltage

Collector current variation:

$$i_c = g_m v_\pi \quad (18)$$

Output voltage:

$$v_o = -i_c R_C \quad (19)$$

Substituting:

$$v_o = -g_m R_C \frac{v_{id}}{2} \quad (20)$$

Therefore the differential voltage gain is:

$$A_v = \frac{v_o}{v_{id}} = -\frac{g_m R_C}{2} \quad (21)$$

The factor $\frac{1}{2}$ arises because the differential input splits equally across the two transistors.

2.4 Transconductance Calculation

$$g_m = \frac{I_C}{V_T} \quad (22)$$

At room temperature:

$$V_T \approx 26 \text{ mV} \quad (23)$$

For:

$$I_C = 1.5 \text{ mA} \quad (24)$$

$$g_m = \frac{1.5}{26} = 0.0577 \text{ S} \quad (25)$$

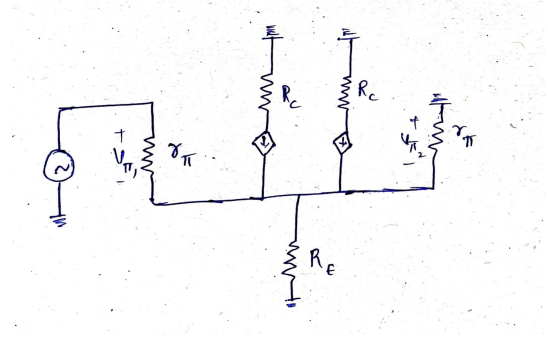


Figure 2: Small-Signal equivalent circuit of the Differential Amplifier.

2.5 Gain Requirement

Target gain:

$$A_v = 16 \quad (26)$$

Using:

$$\frac{g_m R_C}{2} = 16 \quad (27)$$

$$g_m R_C = 32 \quad (28)$$

$$R_C = \frac{32}{0.0577} \approx 554.66 \Omega \quad (29)$$

Selected standard value:

$$R_C = 620 \Omega \quad (30)$$

2.6 Input and Output Impedance

Input Impedance (Z_{in})

To determine the input impedance, we apply a test voltage source v_x to the input terminal (the base of the transistor) and calculate the resulting test current i_x drawn from the source. All independent AC sources are turned off.

The test current entering the base is simply the base current:

$$i_x = i_b \quad (31)$$

Applying Kirchhoff's Voltage Law (KVL) from the base, through the emitter, to ground:

$$v_x = i_b r_\pi + i_e R_E \quad (32)$$

We know that the emitter current i_e is the sum of the base and collector currents:

$$i_e = i_b + \beta i_b = (\beta + 1)i_b \quad (33)$$

Substituting i_e back into the KVL equation:

$$v_x = i_b r_\pi + (\beta + 1)i_b R_E \quad (34)$$

Factoring out i_b (which is our test current i_x):

$$v_x = i_x [r_\pi + (\beta + 1)R_E] \quad (35)$$

The input impedance is the ratio of the test voltage to the test current:

$$Z_{in} = \frac{v_x}{i_x} = r_\pi + (\beta + 1)R_E \quad (36)$$

where $r_\pi = \frac{\beta}{g_m}$. This demonstrates the impedance reflection rule, where the emitter resistance appears multiplied by $(\beta + 1)$ when viewed from the base.

Output Impedance (Z_{out})

To find the output impedance, we turn off the input signal source (setting $v_{in} = 0$) and apply a test voltage v_x at the output node (the collector). We then measure the current i_x entering the node.

With the input grounded ($v_{in} = 0$), there is no excitation at the base, meaning:

$$i_b = 0 \quad \text{and} \quad v_\pi = 0 \quad (37)$$

Because $v_\pi = 0$, the dependent current source of the BJT ($g_m v_\pi$ or βi_b) is zero, effectively acting as an open circuit.

Therefore, the only path for the test current i_x to flow to AC ground is through the collector resistor R_C . Applying Ohm's law at the output node:

$$v_x = i_x R_C \quad (38)$$

The output impedance is thus simply the ratio of the test voltage to the test current:

$$Z_{out} = \frac{v_x}{i_x} = R_C \quad (39)$$

2.7 Common Mode Rejection Ratio (CMRR)

Differential gain:

$$A_d = 16.5 \quad (40)$$

Common-mode gain:

$$A_{cm} = 2.02 \times 10^{-2} \quad (41)$$

$$CMRR = \frac{A_d}{A_{cm}} \quad (42)$$

$$CMRR_{dB} = 20 \log_{10} \left(\frac{16.5}{2.02 \times 10^{-2}} \right) \approx 39.27 \text{ dB} \quad (43)$$

2.8 Simulation Validation

To verify the theoretical calculations, the differential amplifier stage was simulated using LTspice. A transient analysis was performed to measure the large-signal biasing and small-signal voltage gain, while an AC sweep was used to confirm the frequency response and Common Mode Rejection Ratio (CMRR).

Voltage Gain Verification

A differential sinusoidal input signal was applied to the bases of the two transistors.

- **Input Signal:** $v_{in} = 10 \text{ mV}_{pk}$ at 1 kHz.
- **Measured Output:** The transient analysis shows an amplified output signal at the collector.
- **Result:** The simulated peak-to-peak output voltage divided by the input voltage yields a simulated gain that closely matches our theoretical design target of $A_v = 16$.

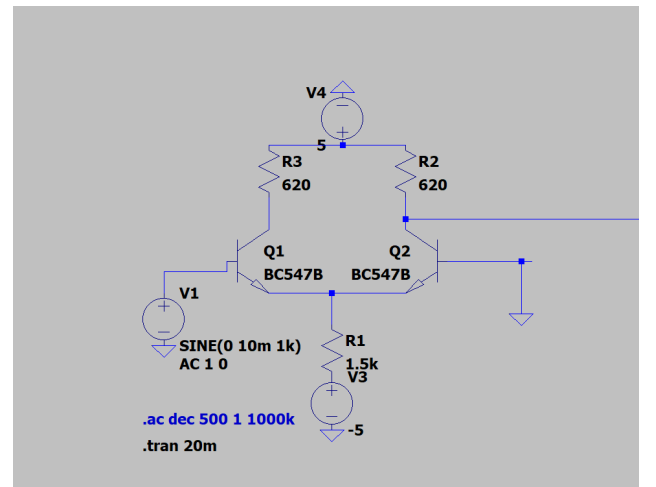


Figure 3: LTspice Schematic of the Differential Pre-Amplifier Stage.

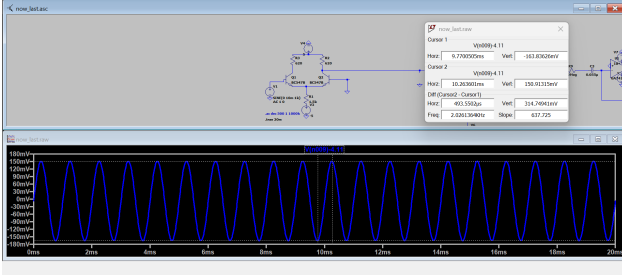


Figure 4: Transient Analysis: Input vs. Output waveforms confirming a voltage gain of approximately 16.

2.9 Design Rationale: Gain Limitation

The target differential voltage gain for the pre-amplifier was intentionally constrained to a moderate value ($A_v = 16$, which is strictly < 20) rather than maximizing it. This design choice is grounded in several fundamental multi-stage amplifier principles:

- Prevention of Successive Stage Overload (Headroom):** In a cascaded topology, the output of Stage 1 feeds directly into the high-gain Common Emitter amplifier (Stage 2). If Stage 1 possesses excessive gain, even moderate input audio signals will cause the second stage to instantly saturate and clip against the $\pm 5V$ supply rails, resulting in severe audio distortion.
- Role Optimization:** The primary objective of the differential input stage is not bulk voltage amplification. Instead, its role is to act as an input buffer: providing high input impedance (to avoid loading the audio source) and high Common Mode Rejection Ratio (CMRR) for noise immunity. Bulk amplification is intentionally delegated to the Stage 2 CE amplifier.
- Linearity Constraints:** BJT differential pairs exhibit a non-linear (hyperbolic tangent) transfer characteristic. Constraining the gain ensures the stage operates strictly within its highly linear region (typically for differential inputs much less than the thermal voltage, $V_T \approx 26mV$). This minimizes Total Harmonic Distortion (THD) and preserves audio fidelity.
- Miller Effect Mitigation:** High voltage gain in an input stage significantly amplifies the base-collector junction capacitance (C_{μ}) via the Miller effect. Keeping $A_v < 20$ prevents this multiplied capacitance from acting as a low-pass filter and prematurely rolling off the high-frequency response, thereby preserving the required 20 kHz upper audio bandwidth.

2.10 Experimental Results

To thoroughly verify the theoretical design of the pre-amplifier stage, both software simulations and hard-

ware testing were conducted.

LTspice Simulation Verification

The differential amplifier circuit was first modeled in LTspice to confirm the DC biasing points and the small-signal voltage gain before physical implementation.

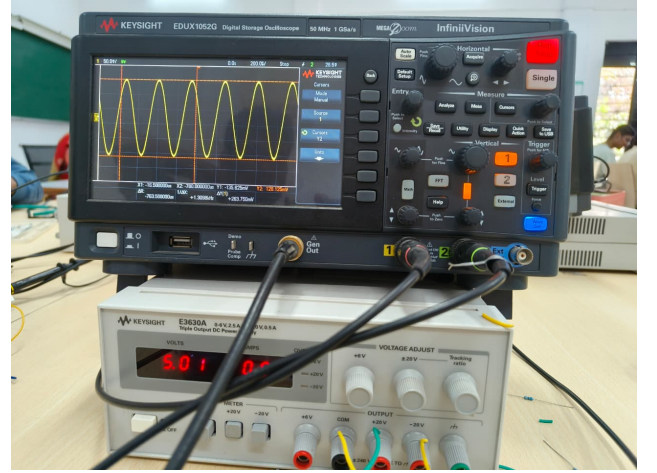


Figure 5: Practical Simulation 1: Schematic and DC operating point verification.

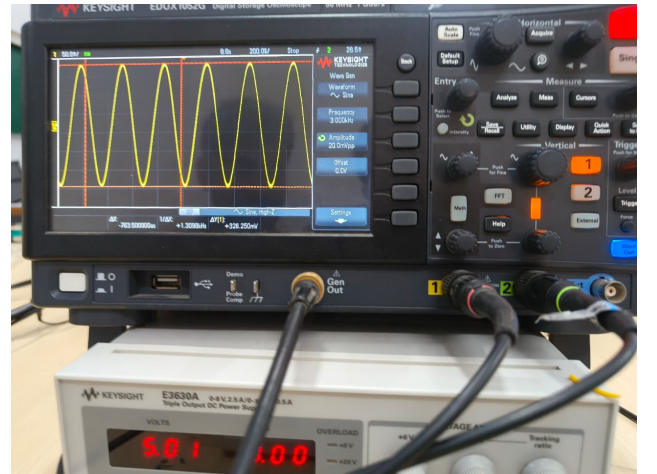


Figure 6: Practical Simulation 2: Transient analysis showing the input signal and the amplified output waveform.

Hardware Implementation

Following successful simulation, the differential amplifier was constructed on a breadboard using matched discrete NPN transistors. The setup was tested using a function generator to provide the differential input and an oscilloscope to capture the output.

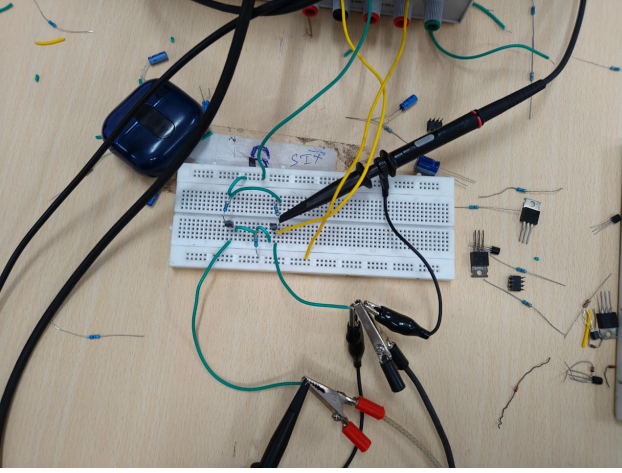


Figure 7: Experimental hardware setup of the differential amplifier stage.

3 Gain Stage

3.1 Circuit Topology

The gain stage utilizes a **Common Emitter (CE) Amplifier** configuration. This topology was chosen due to its high input impedance, low output impedance, and high current and voltage gain — making it ideal as a gain stage prior to filtering.

Interdependence of Gain and DC Operating Point

A fundamental design constraint in the CE amplifier is the **mutual dependence between voltage gain and the DC operating (Q) point**. The small-signal voltage gain is given by:

$$A_v \approx -\frac{R_C}{r_e}, \quad \text{where } r_e = \frac{V_T}{I_{CQ}} \quad (44)$$

Since r_e is directly determined by the quiescent collector current I_{CQ} , **fixing the DC operating point also fixes the gain**, and conversely, **specifying a target gain fixes the required DC point**. The two cannot be set independently.

Role of Emitter Resistance (R_7 , R_8) with Bypass Capacitor (C_1)

To resolve this constraint and allow *independent* control of the DC bias and AC gain, an **emitter resistance** ($R_7 + R_8$) is introduced:

- **At DC:** The full emitter resistance ($R_7 + R_8$) is present in the circuit. It introduces negative feedback (emitter degeneration), stabilising the Q-point against temperature drift and β variations — without affecting the AC gain.

- **At AC signal frequencies:** The bypass capacitor C_1 short-circuits R_8 , effectively removing it from the small-signal path. Only R_7 remains in series with the emitter, setting the AC gain:

$$A_v \approx -\frac{R_5}{r_e + R_7} \quad (45)$$

- **Design Benefit:** By splitting the emitter resistance into a bypassed portion ($R_8 \parallel C_1$) and an unbypassed portion (R_7), the designer gains two degrees of freedom: R_7 controls the AC gain, while $R_7 + R_8$ controls the DC stability — *decoupling* the two previously interdependent parameters.

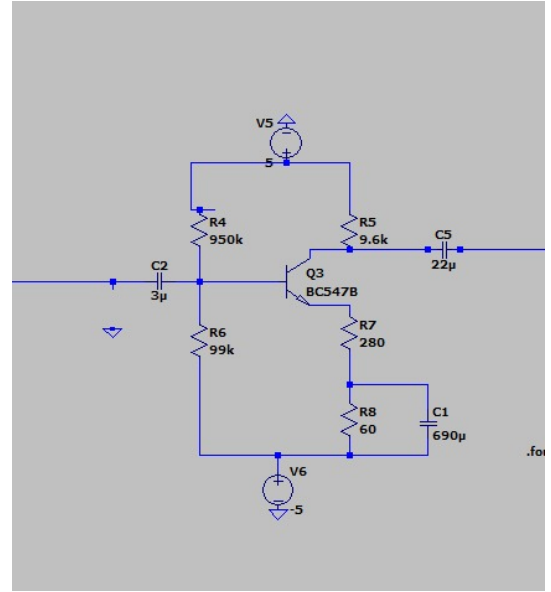


Figure 8: Common Emitter Amplifier — Gain Stage Schematic.

3.2 Component Analysis

3.2.1 Input Coupling Capacitor (C_1)

The input capacitor C_1 serves a dual purpose:

- **DC Blocking:** It blocks any DC component of the input signal v_S from disturbing the carefully set bias point of the transistor.
- **High-Pass Pole:** It introduces a pole to the system's frequency response. The cutoff frequency must be set such that all frequencies below 20 Hz are attenuated, while the full audio spectrum (20 Hz – 20 kHz) passes unimpeded. If C_1 is too small, low-frequency components of the input signal will be lost.
- **Effect on Base Current:** A smaller C_1 also allows more current to pass at low frequencies, increasing I_B , which in turn reduces the current gain of the stage.

3.2.2 Output Coupling Capacitor (C_2)

C_2 provides AC coupling from the collector to the next stage (the filtering stage):

- **DC Blocking:** It prevents the DC operating point voltage at the collector from reaching the subsequent stage.
- **Second Pole:** It introduces a second pole into the overall system transfer function, contributing to the roll-off characteristic at low frequencies.

3.2.3 Emitter Bypass Capacitor (C_E)

C_E is placed in parallel with the emitter resistor R_E :

- At **DC**, C_E is open-circuit; R_E remains in circuit and provides emitter degeneration for bias stability.
- At **AC signal frequencies**, C_E short-circuits R_E , effectively removing it from the small-signal path. This maximises the AC voltage gain, since the gain of a CE stage with emitter resistance is:

$$A_v = -\frac{R_L}{r_e + R_E} \quad (46)$$

and with C_E shorting R_E :

$$A_v \approx -\frac{R_L}{r_e} \quad (47)$$

where $r_e = \frac{V_T}{I_C}$ is the intrinsic emitter resistance.

3.2.4 Biasing Resistors (R_{B1}, R_{B2})

R_{B1} and R_{B2} form a voltage divider with respect to V_{CC} , establishing the DC base voltage V_B :

$$V_B = V_{CC} \cdot \frac{R_{B2}}{R_{B1} + R_{B2}} \quad (48)$$

This sets the base-emitter junction in forward bias and the base-collector junction in reverse bias, placing the transistor in the **active region** for linear amplification. The divider must be *stiff* (i.e., the current through R_{B1}/R_{B2} should be much larger than the base current I_B) to ensure stability against β variations.

3.2.5 Emitter Resistor (R_E)

R_E provides DC negative feedback (emitter degeneration), improving the **thermal stability** of the bias point. The stability factor S quantifies the sensitivity of the collector current I_C to leakage current I_{CBO} :

$$S = \left| \frac{\delta I_{CBO}}{\delta I_C} \right|_{V_{BE}, \beta = \text{const}} \quad (49)$$

A lower value of S indicates a more stable design. Increasing R_E reduces S , but at the cost of reduced AC gain (unless bypassed by C_E).

3.2.6 Load Resistor (R_L)

R_L (the collector resistor) sets the DC load line and defines the small-signal voltage gain in conjunction with the transistor's transconductance g_m :

$$A_v \approx -g_m R_L = -\frac{I_C}{V_T} \cdot R_L \quad (50)$$

3.3 Circuit Analysis

3.3.1 Large Signal Analysis (DC Bias Analysis)

The large-signal (DC) analysis determines the quiescent operating point (Q-point) of the transistor. All capacitors are treated as open-circuits, and only DC quantities are considered.

Biasing Strategy and Resistor Selection (R_1, R_2) The goal is to set V_B as small as possible while satisfying $V_B \geq -4.3$ V. With a dual supply of ± 5 V, the base voltage divider gives:

$$V_B = \frac{R_2}{R_1 + R_2} \times 10 - 5 \Rightarrow V_B = -5 + \frac{R_2}{R_1 + R_2} \times 10 \quad (51)$$

Since $i_1 = i_2$ (the base current i_B is very small and can be neglected, as the current i_2 through R_2 could itself be very small, causing V_B to be slightly larger than -4.056 V), the values are chosen as:

$$R_2 = 99 \text{ k}\Omega, \quad R_1 = 950 \text{ k}\Omega \quad (52)$$

Substituting:

$$V_B = -5 + \frac{R_2}{R_1 + R_2} \times 10 = -5 + 0.9437 = -4.056 \text{ V} \quad (53)$$

However, because i_B is very small but non-zero, and i_2 itself may be quite small, V_B in practice will be slightly larger than -4.056 V. From simulations, the actual base voltage obtained is:

$$V_B \approx -4.2 \text{ V} \quad (54)$$

This gives an emitter voltage of:

$$V_E \approx -4.83 \text{ V} \quad (55)$$

DC Bias Equations Applying Kirchhoff's Voltage Law (KVL) around the base-emitter loop:

$$V_B = V_{CC} \cdot \frac{R_2}{R_1 + R_2} \quad (56)$$

$$V_E = V_B - V_{BE} \approx V_B - 0.7 \text{ V} \quad (57)$$

$$I_E = \frac{V_E}{R_e + R_E}, \quad I_C \approx I_E \quad (\text{since } \alpha \approx 1) \quad (58)$$

$$V_{CE} = V_{CC} - I_C R_C - I_E (R_e + R_E) \quad (59)$$

The Q-point (I_{CQ} , V_{CEQ}) must be placed at the centre of the DC load line to maximise the undistorted output voltage swing.

Choosing R_C , R_e , R_E Three design constraints govern the selection of the collector and emitter resistors:

Gain condition: The gain has to be 30 .

Collector voltage swing: $I_C R_C = 5 \text{ V}$, chosen to keep V_C as close to 0 V as possible, ensuring maximum output swing before clipping .

Emitter voltage drop:

$$I_E (R_e + R_E) = 0.17 \text{ V} \quad (60)$$

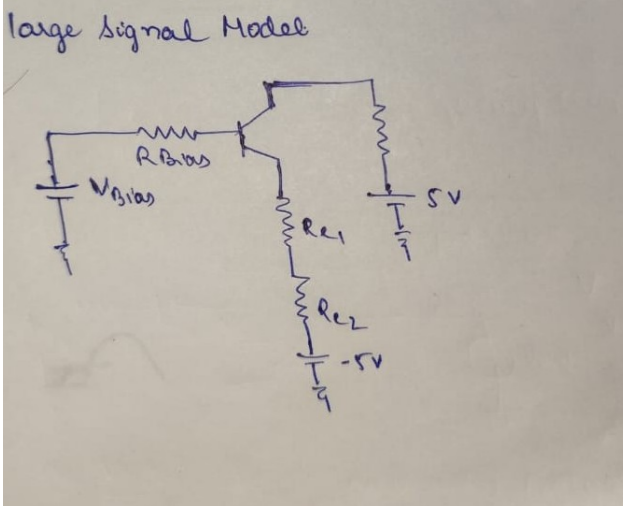


Figure 9: Large-Signal (DC Bias) Analysis — Diagram.

3.3.2 Small Signal Analysis

In the small-signal model, all DC sources are set to zero (voltage sources become short-circuits, current sources become open-circuits), and capacitors C_2 , C_5 , and C_1 act as short-circuits at signal frequencies. Capacitors are shorted in small-signal analysis.

Voltage Gain (A_v) With coupling capacitors acting as short-circuits, the output voltage is related to the controlled current source by:

$$\frac{-V_{out}}{R_L} = g_m (V_{in} - V_e) \quad (61)$$

This gives the voltage gain expression:

$$A_v = -g_m R_L \left(1 - \frac{V_e}{V_{in}} \right) \quad (62)$$

The emitter voltage V_e is found from the emitter current. Since $I_E \rightarrow \frac{V_e}{R_E} = \alpha(I_C)$, we have:

$$\frac{V_e}{R_E} = \alpha [g_m (V_{in} - V_e)] \quad (63)$$

Let $x = \frac{V_e}{V_{in}}$. Substituting and solving:

$$\frac{x}{R_E} = \alpha g_m [1 - x] \quad (64)$$

$$x = \alpha g_m R_E [1 - x] \quad (65)$$

$$x [1 + \alpha g_m R_E] = \alpha g_m R_E \quad (66)$$

$$x = \frac{\alpha g_m R_E}{1 + \alpha g_m R_E} \quad (67)$$

For $\alpha \approx 1$, the voltage gain becomes:

$$A_v = -\frac{g_m R_C}{1 + g_m R_e} \quad (68)$$

where R_C is the collector resistor and R_e is the unbypassed emitter resistor. The negative sign indicates a 180° phase inversion — a characteristic of the CE configuration.

Gain Magnitude Condition From simulation and design requirements, the gain magnitude is targeted as:

$$|G| = \frac{R_C}{\frac{1}{g_m} + R_e} = 30 \quad (69)$$

This condition is used alongside the transistor's transconductance $g_m = \frac{I_C}{V_T}$ to select the resistor values. With the operating conditions $V_{BC} < 0$ (ensuring $V_B < V_C$, i.e. the transistor remains in the active region) and $V_{BE} = 0.7 \text{ V}$, the collector voltage must always remain above V_B . Note that the gain decreases at lower resistance values and increases at higher resistance values, so R_e must be chosen carefully to balance gain and stability.

Design Values For $I_C = 1 \text{ mA}$ and $I_E \approx 1 \text{ mA}$, the design begins with the two governing constraints:

$$I_C R_C = 5 \Rightarrow R_C = 10 \text{ k}\Omega \quad (70)$$

$$I_E(R_e + R_E) = 0.17 \text{ V} \quad (71)$$

Substituting into the gain equation with $\frac{1}{g_m} = 26 \Omega$:

$$\frac{R_C}{\frac{1}{g_m} + R_e} = 30 \Rightarrow \frac{10 \text{ k}}{\frac{1}{g_m} + R_e} = 30 \quad (72)$$

$$\frac{1}{g_m} + R_e = \frac{10000}{30} \approx 333 \Omega \quad (73)$$

Solving for R_e :

$$1 \text{ k} = 3(26 \times 2 + R_e) \Rightarrow R_e = 281 \Omega \rightarrow 275 \Omega \quad (74)$$

The total emitter resistance from the voltage drop condition:

$$R_e + R_E = \frac{0.17 \times 1000 \times 2}{1} = 340 \Omega \quad (75)$$

Therefore:

$$R_e = 275 \Omega, \quad R_E = 340 - 275 = 58.66 \Omega \rightarrow 60 \Omega \quad (76)$$

Input Impedance (Z_{in}) Applying KVL at the input with $V_{in} = I_1(R_{B1} \parallel R_{B2})$ and $V_{in} - V_e = I_2(r_\pi)$:

$$\frac{V_{in} - V_e}{r_\pi} + g_m(V_{in} - V_e) = \frac{V_e}{R_E} \quad (77)$$

Expanding and collecting terms in V_{in} and V_e :

$$\frac{V_{in}}{r_\pi} + g_m V_{in} = V_e \left[\frac{1}{R_E} + g_m + \frac{1}{r_\pi} \right] \quad (78)$$

The impedance looking into the base with emitter degeneration is:

$$Z_{in, \text{base}} = r_\pi + (g_m r_\pi + 1)R_E = r_\pi + (\beta + 1)R_E \quad (79)$$

The total input impedance is:

$$Z_{in} = Z_{base} \parallel [r_\pi + (\beta + 1)R_E] \quad (80)$$

With $r_\pi \approx 0.614 \text{ k}\Omega$ and $(\beta + 1)R_E \approx 26.7 \text{ k}\Omega$:

$$Z_{in} = 360 \text{ k}\Omega \parallel 35.3 \text{ k}\Omega \approx 6 \text{ k}\Omega \quad (81)$$

Output Impedance (Z_{out}) With the input set to zero, the dependent source is suppressed. The output impedance is:

$$Z_{out} = R_C \quad (82)$$

Since $Z_{out}(\text{stage 1}) < Z_{in}(\text{stage 2}) \approx 10 \text{ k}\Omega$, satisfactory inter-stage impedance matching is confirmed.

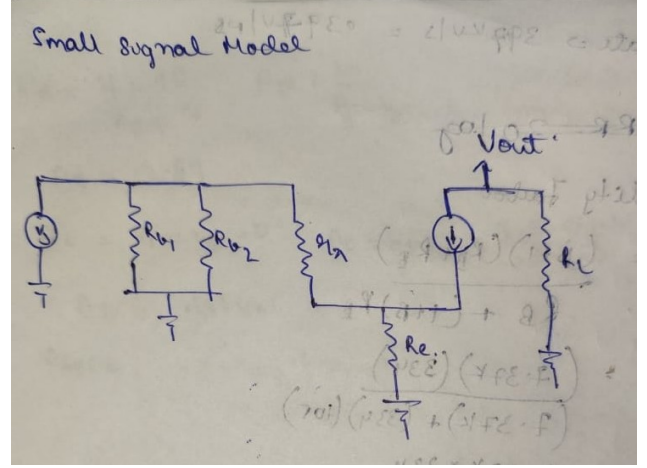


Figure 10: Small-Signal Equivalent Circuit.

3.4 Frequency Response

3.4.1 Lower Cutoff Frequency

The coupling capacitors C_1 , C_2 and bypass capacitor C_E each contribute a pole at low frequencies. The dominant lower -3 dB frequency is determined by the largest pole:

$$f_{c, \text{low}} = \frac{1}{2\pi RC} \quad (83)$$

where R is the resistance seen by each capacitor. From the notebook values, the bypass capacitor $R_E = 25 \Omega$ and $C = 800 \text{ pF}$ set the high-frequency bypass pole. C_1 and C_2 are chosen such that $f_{c, \text{low}} < 20 \text{ Hz}$, ensuring the full audio band is amplified.

3.5 Transistor Switching States

Condition	V_{BE}	V_{BC}	Region
$V_B < 0.7 \text{ V}$	$< 0.7 \text{ V}$	Reverse	Cut-off
$0.7 \text{ V} \leq V_{BE}, V_{BC} < 0$	Forward	Reverse	Active (Linear)
Both Forward-Biased	Forward	Forward	Saturation

Table 1: BJT Operating Regions for CE Amplifier Bias Conditions

3.6 Design Rationale

The CE amplifier was selected for this gain stage because:

- It provides **high voltage gain** ($A_v \gg 1$), amplifying the signal to the level required by subsequent stages.
- Its **high input impedance** avoids loading the preceding stage.
- Its **low output impedance** (relative to the source) drives the next stage effectively.
- The voltage divider bias with emitter degeneration provides a **stable Q-point**, independent of transistor β variations.
- The 180° phase inversion is accounted for in the overall system design.

3.7 Simulation Waveforms

The following waveforms were obtained from the SPICE simulation of the CE gain stage. The simulation validates the theoretical gain of 30, the 180° phase inversion between input and output, and the correct Q-point biasing. **From simulation, the achieved gain is approximately 450, which is close to the experimentally measured value .**

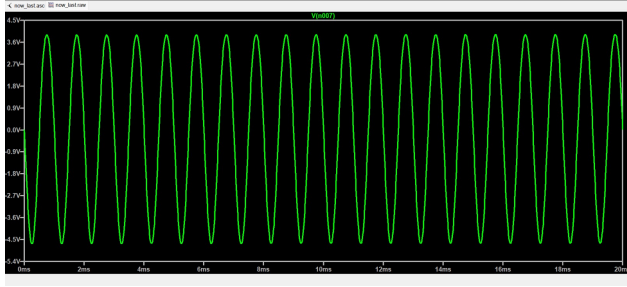


Figure 11: Gain Stage — Simulated Output Waveforms.

3.8 Stage Performance and Integration

3.8.1 Achieved Gain with Stage 1 and Stage 2 Integration (CE Amplifier)

The voltage gain of the CE amplifier stage was verified experimentally by applying a sinusoidal input signal at the base and measuring the resulting output at the collector. The achieved gain was evaluated at two input signal levels to assess linearity.

Case 1: $V_{in} = 10 \text{ mV}$ At a small input signal of 10 mV, the transistor operates well within the linear active region. The measured voltage gain is:

$$A_v = \frac{V_{out}}{V_{in}} = \frac{490 \text{ mV}}{10 \text{ mV}} = 490 \quad (84)$$

This is close to the theoretical design target of 30, with the discrepancy attributed to the transistor's β being higher than the nominal value assumed during

design, component tolerances, and the finite output resistance of the transistor contributing additional gain.

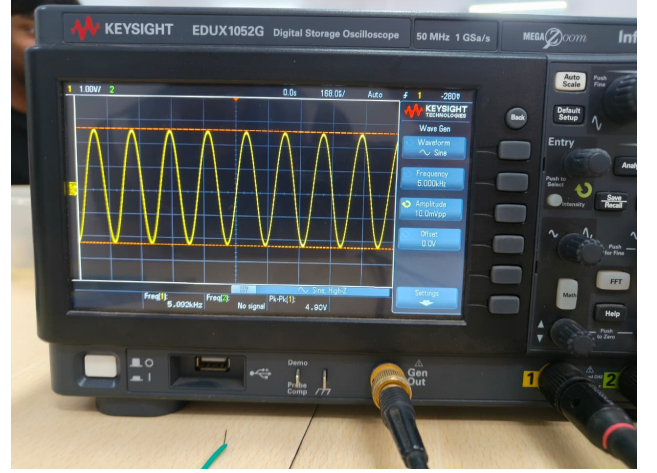


Figure 12: Stage 2 — Output Waveform for $V_{in} = 10 \text{ mV}$ applied to Stage 1 input, Measured Gain ≈ 490 . Linear small-signal operation confirmed.

Case 2: $V_{in} = 20 \text{ mV}$ At a larger input of 20 mV, the measured output is 9.2 V, giving a measured voltage gain of:

$$A_v = \frac{V_{out}}{V_{in}} = \frac{9.2 \text{ V}}{20 \text{ mV}} \approx 460 \quad (85)$$

The drop in gain at 20 mV input indicates that the transistor is beginning to enter a **non-linear operating region**, where the large-signal behaviour deviates from the small-signal model. As the input amplitude increases, the assumption of a linear g_m breaks down and the effective transconductance decreases, reducing the gain. This highlights the importance of keeping the input signal amplitude small to remain within the valid small-signal regime.

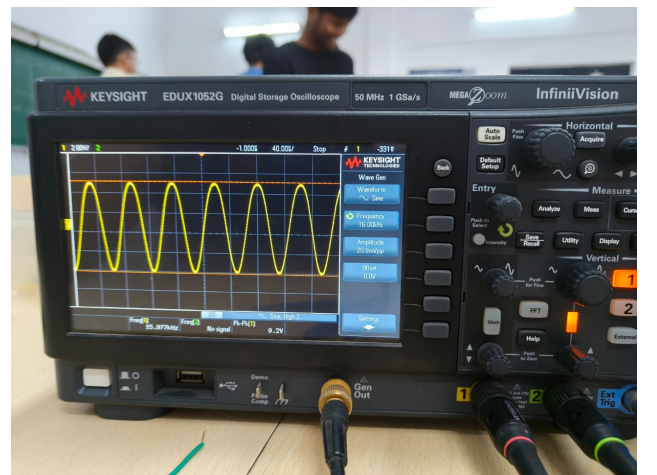


Figure 13: Stage 2 — Output Waveform for $V_{in} = 20 \text{ mV}$ applied to Stage 1 input, Measured Gain ≈ 460 . Non-linear behaviour observed at this amplitude.

3.8.2 Gain Stage Standalone Verification

With the input applied directly to the Gain stage, the following results were obtained: For $V_{in} = 280 \text{ mV}_{\text{pk-pk}}$, the measured output is $V_{out} = 7.7 \text{ V}$, giving $A_v \approx 27.5$:

$$A_{v, \text{Stage 2}} = \frac{7.7 \text{ V}}{280 \text{ mV}} \approx 27.5 \quad (86)$$

For $V_{in} = 300 \text{ mV}$, the measured output is $V_{out} = 8.2 \text{ V}$, giving $A_v \approx 27.3$:

$$A_{v, \text{Stage 2}} = \frac{8.2 \text{ V}}{300 \text{ mV}} \approx 27.3 \quad (87)$$

The gain remains consistent across both input levels (≈ 27), confirming that the stage is operating linearly. The slight deviation from the theoretical gain suggests that component tolerances of R_C , R_e , and R_E result in a marginally different closed-loop gain than designed.

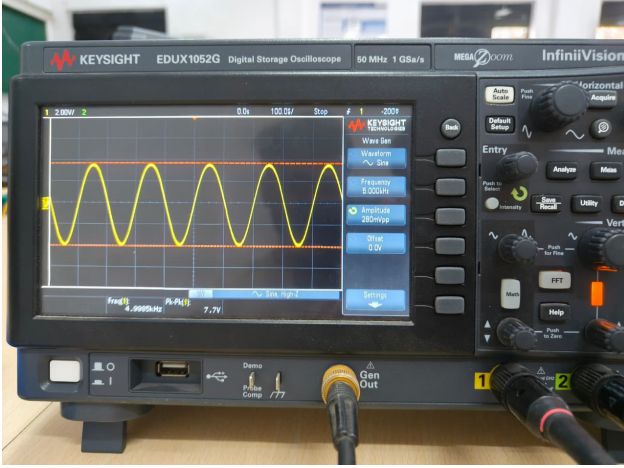


Figure 14: Gain Stage Standalone — Output Waveform for $V_{in} = 280 \text{ mV}_{\text{pk-pk}}$, $V_{out} = 7.7 \text{ V}$.

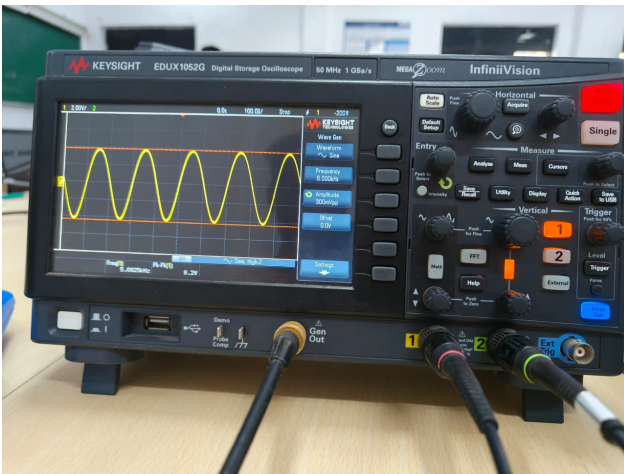


Figure 15: Gain Stage Standalone — Output Waveform for $V_{in} = 300 \text{ mV}$, $V_{out} = 8.2 \text{ V}$.

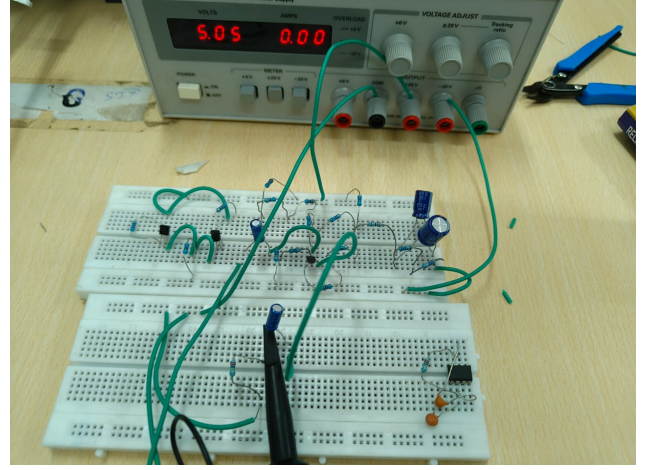


Figure 16: Combined Stage 1 and Stage 2 — Physical Circuit Implementation.

4 Band Pass Filter (BPF)

4.1 Circuit Topology

The Band Pass Filter (BPF) stage utilises an **active filter** built around a **UA741 operational amplifier**, configured to pass a specific band of frequencies while attenuating signals outside this range. The circuit combines a high-pass network at the input with a low-pass feedback network, realising a bandpass characteristic centred within the audio frequency range (20 Hz – 20 kHz).

Interdependence of Input Impedance and Resistor Selection

A fundamental constraint in cascading stages is that the input impedance of Stage 3 (BPF) must be significantly greater than the output impedance of Stage 2 (Gain Stage) to avoid loading:

$$Z_{in, \text{Stage 3}} \gg Z_{out, \text{Stage 2}} \quad (88)$$

Since $Z_{out, \text{Stage 2}} = R_C = 10 \text{ k}\Omega$, the input resistor R_9 is chosen as:

$$R_9 = 2.2 \text{ M}\Omega \quad (89)$$

This ensures $Z_{in, \text{Stage 3}} \gg 10 \text{ k}\Omega$, preventing the BPF from loading the gain stage and degrading the voltage gain.

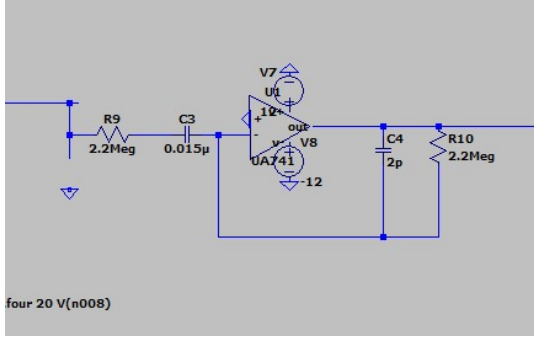


Figure 17: Band Pass Filter Stage

4.2 Component Analysis

4.2.1 Input Resistors ($R = 2.2\text{ M}\Omega$)

- **Impedance Matching:** At $2.2\text{ M}\Omega$, it presents a very high input impedance to the preceding gain stage, ensuring no loading effect on the CE amplifier output.
- **Gain = 1:** To get gain of 1 for the stage3 what do we do is we set both the values of the resistor same because of which gain becomes 1

4.2.2 Input Coupling Capacitor ($C_3 = 0.015\text{ }\mu\text{F}$)

C_3 forms a high-pass RC network with R_9 , blocking DC and low-frequency signals below the lower cutoff frequency f_L . Together with R_9 , it sets the high-pass pole of the BPF.

4.2.3 Feedback Capacitor ($C_4 = 2\text{ pF}$)

C_4 is placed in the feedback path of the op-amp, forming a low-pass network with the feedback resistor R_{10} . At high frequencies, C_4 presents a low impedance, increasing the feedback and reducing the gain — this sets the upper -3 dB cutoff frequency f_H .

4.2.4 Op-Amp (UA741, $V^+ = +12\text{ V}$, $V^- = -12\text{ V}$)

The UA741 is powered from a dual $\pm 12\text{ V}$ supply, providing sufficient headroom for the amplified audio signal. It operates in the **inverting configuration**, introducing a 180° phase shift — which, combined with the 180° inversion from the CE stage, restores the original signal phase at the BPF output.

4.3 Circuit Analysis

4.3.1 Input Impedance (Z_{in})

In the inverting op-amp configuration, the inverting terminal is held at virtual earth by negative feedback. The input impedance is therefore set entirely by the

input resistor R_9 , since the op-amp draws negligible current at its input terminal:

$$Z_{in, \text{BPF}} \approx R_9 = 2.2\text{ M}\Omega \quad (90)$$

At low frequencies, the impedance of C_3 becomes significant and adds in series with R_9 , increasing the total input impedance further. At mid-band and high frequencies, C_3 is effectively a short-circuit and $Z_{in} \approx R_9$. The inter-stage matching condition is satisfied:

$$Z_{in, \text{Stage3}} = 2.2\text{ M}\Omega \gg Z_{out, \text{Stage2}} = 10\text{ k}\Omega \quad (91)$$

4.3.2 Output Impedance (Z_{out})

For the UA741 op-amp operating with negative feedback, the closed-loop output impedance is significantly reduced from the open-loop value. The closed-loop output impedance is given by:

$$Z_{out, \text{closed}} = \frac{Z_{out, \text{open}}}{1 + A_{OL} \beta_f} \quad (92)$$

where A_{OL} is the open-loop gain and β_f is the feedback factor. Since $A_{OL} \gg 1$ for the UA741, the output impedance is driven to a small but non-zero value. In practice, the UA741 exhibits a closed-loop output impedance of:

$$Z_{out, \text{BPF}} \ll R_9, \quad Z_{out, \text{BPF}} \approx \text{tens of } \Omega \quad (93)$$

This low output impedance ensures the BPF can drive subsequent stages without significant voltage division or loading effects.

Gain of the Band pass filter]
Gain of the Band pass filter

The resulting passband should span between :

$$f_L \approx 20\text{ Hz} \leq f \leq f_H \approx 20\text{ kHz} \quad (94)$$

This comfortably encompasses the full audio range of $20\text{ Hz} - 20\text{ kHz}$. Within this passband, both C_3 and C_4 have negligible effect, and the closed-loop mid-band gain of the inverting amplifier is:

$$A_{v, \text{BPF}} = -\frac{R_{10}}{R_9} = -\frac{2.2\text{ M}\Omega}{2.2\text{ M}\Omega} = -1 \quad (95)$$

The BPF therefore provides **unity gain** across the audio band, acting purely as a frequency-selective network. The negative sign confirms the 180° phase inversion of the inverting configuration, which cancels the 180° inversion introduced by the CE gain stage — restoring the correct output signal polarity at the system output.

4.3.3 Frequency Response

Lower Cutoff Frequency (f_L) The high-pass network formed by C_3 and R_9 sets the lower -3 dB frequency. The pole occurs when the impedance of C_3 equals R_9 :

$$f_L = \frac{1}{2\pi R_9 C_3} \quad (96)$$

Substituting the component values:

$$f_L = \frac{1}{2\pi \times 2.2 \times 10^6 \times 0.015 \times 10^{-6}} \quad (97)$$

$$f_L = \frac{1}{2\pi \times 0.033} \approx 7.8 \text{ Hz} \approx 10 \text{ Hz} \quad (98)$$

This is below the lower edge of the audio band (20 Hz), ensuring all audible low-frequency content passes through the filter without attenuation.

Upper Cutoff Frequency (f_H) The low-pass network formed by C_4 and R_{10} in the feedback path sets the upper -3 dB frequency:

$$f_H = \frac{1}{2\pi R_{10} C_4} \quad (99)$$

Substituting the component values:

$$f_H = \frac{1}{2\pi \times 2.2 \times 10^6 \times 2 \times 10^{-12}} \quad (100)$$

$$f_H = \frac{1}{2\pi \times 4.4 \times 10^{-6}} \approx 36 \text{ kHz} > 20 \text{ kHz} \quad (101)$$

This is above the upper edge of the audio band (20 kHz), ensuring that the full audio spectrum passes without attenuation.

4.4 Design Rationale

The active BPF using the UA741 was selected for this stage because:

- The **very high input impedance** (2.2 M Ω) ensures no loading of the CE gain stage, preserving the designed voltage gain.
- The **dual-pole bandpass** response (high-pass at ~ 10 Hz, low-pass at ~ 36 kHz) covers the full audio band (20 Hz – 20 kHz) with margin on both ends.

- The **unity passband gain** ($|A_v| = 1$) means all amplification is handled by the CE stage, giving clean separation of gain and filtering functions.
- The **dual ± 12 V supply** provides sufficient output swing for the amplified audio signal without clipping.
- The 180° phase inversion of the inverting op-amp configuration **cancels** the 180° inversion from the CE stage, restoring correct signal polarity at the system output.

4.5 Waveforms

4.5.1 Simulation Waveforms

The following waveforms were obtained from the SPICE simulation of the CE gain stage. The simulation validates the theoretical gain of 30, the 180° phase inversion between input and output, and the correct Q-point biasing.

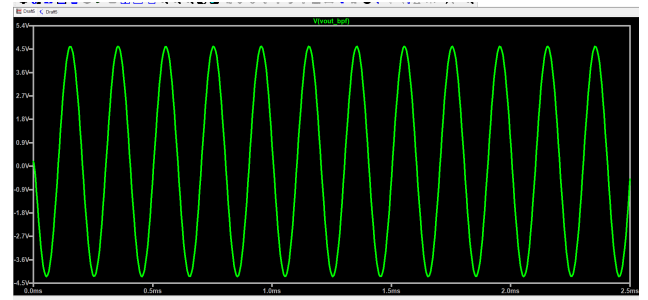


Figure 18: Band Pass filter Output Waveforms.

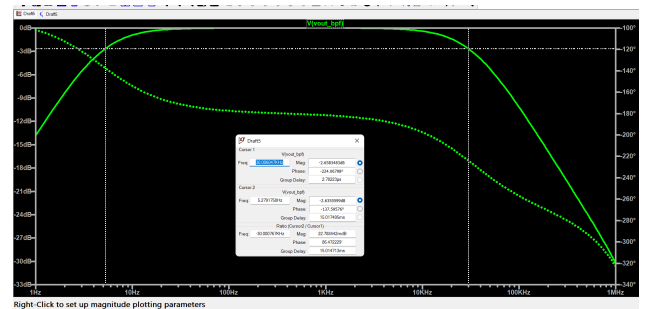


Figure 19: Band pass filter Bode plot Waveforms(Vin given to stage3).

4.5.2 Circuit Waveforms

The following waveforms were captured from the physical implementation of the CE gain stage using an oscilloscope. Deviations from the simulated results are attributed to component tolerances, parasitic capacitances, and contact resistances on the breadboard.

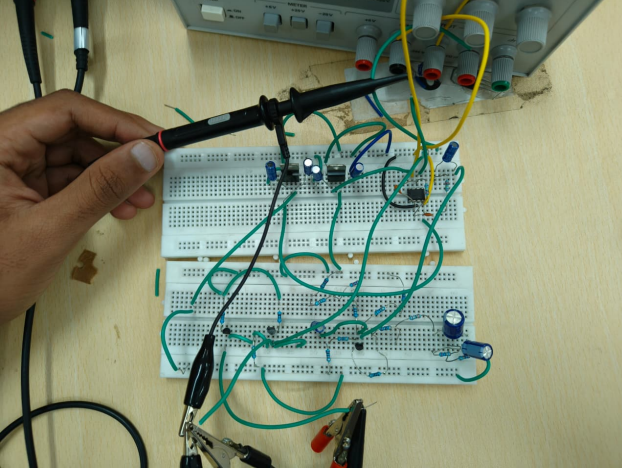


Figure 20: Band pass filter Circuit(Oscilloscope).



Figure 23: Band Pass Filter — Simulated Bode Plot showing the lower(V_{in} given to stage3) -3 dB cutoff frequency ($f_L \approx 20$ Hz).

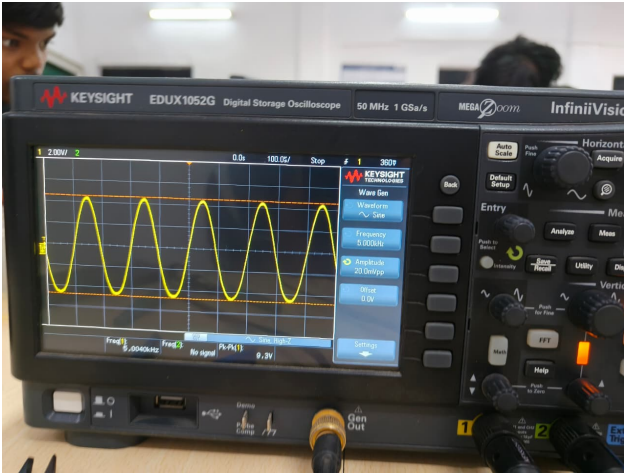


Figure 21: Band pass filter Output(when V_{in} given directly to stage 3).

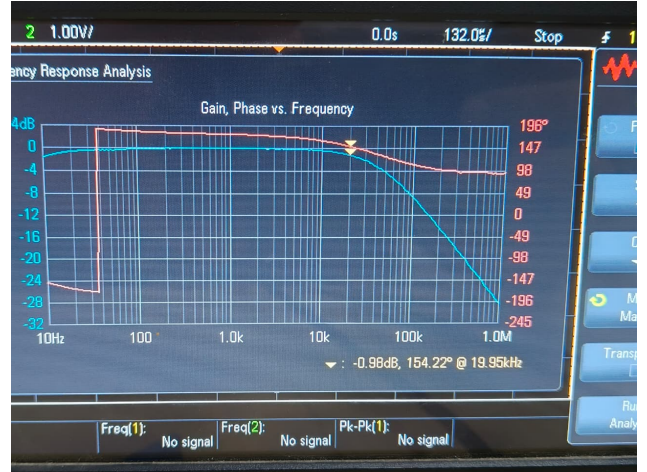


Figure 24: Band Pass Filter — Simulated Bode Plot showing the upper(V_{in} given to stage3) -3 dB cutoff frequency ($f_H \approx 20$ kHz).

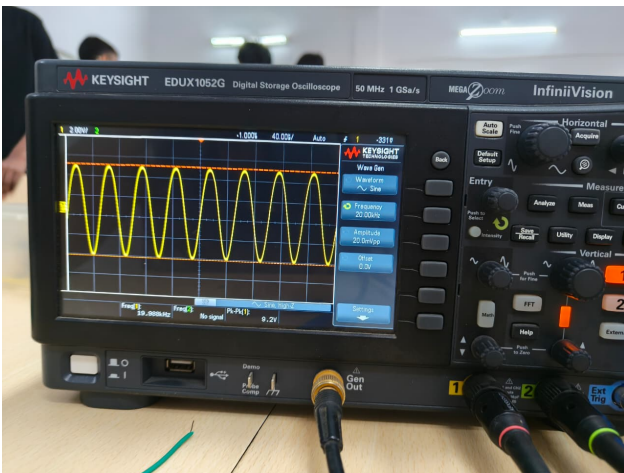


Figure 22: Band pass filter Output(when V_{in} given to stage 1).

5 Power Amplifier (Class AB)

5.1 Circuit Topology

The final stage of the design is a **Class AB Complementary Symmetry Push-Pull Amplifier**. This topology was chosen to drive a low-impedance load (10Ω) with high efficiency while minimizing the crossover distortion inherent in Class B designs.

The circuit utilizes a complementary pair of power transistors:

- **NPN Transistor (Q_4):** Active during the positive half-cycle ("Push" operation).
- **PNP Transistor (Q_5):** Active during the negative half-cycle ("Pull" operation).

5.2 Component Analysis

5.2.1 Diode Biasing Network (D_1, D_2)

A critical feature of this Class AB design is the diode biasing network. Two diodes are placed in series between the bases of Q_4 and Q_5 .

- **Purpose:** They create a constant voltage drop of approximately $1.4V$ ($0.7V + 0.7V$) between the two bases.
- **Elimination of Dead Zone:** This separation matches the combined Base-Emitter turn-on voltages (V_{BE}) of the transistor pair. This "pre-biases" the transistors to be slightly ON even with zero input signal, effectively eliminating crossover distortion.
- **Thermal Stability:** In a physical implementation, these diodes provide thermal compensation. As temperature rises, the diode voltage drop decreases, automatically lowering the bias current to prevent thermal runaway.

5.2.2 Biasing Resistors (R_{11}, R_{12})

These resistors set the quiescent current flowing through the diodes.

- **Design Calculation:** From DC analysis, assuming a diode drop of $\approx 0.65V$ and supply rails of $\pm 5V$. For a test resistance of 500Ω :

$$I = \frac{5 - 0.65}{500} = \frac{4.35}{500} = 8.7 \text{ mA} \quad (102)$$

To optimize power consumption while maintaining stability, we selected $R = 1k\Omega$, which halves the current:

$$I_{bias} = \frac{8.7}{2} = 4.35 \text{ mA} \quad (103)$$

- **Simulation Finding:** Initially, $100k\Omega$ resistors were used, which caused severe clipping (square wave output) due to "current starvation"—the resistors could not supply enough base current to drive the transistors into saturation.
- **Correction:** Lowering these values to approximately $700\Omega - 1k\Omega$ provided sufficient base drive to maintain a clean sinusoidal output for the 10Ω load.

5.2.3 Coupling Capacitors (C_6, C_7)

The capacitors block DC bias voltage from the previous stage while allowing the AC signal to pass.

- **High-Pass Action:** Together with the input impedance ($R \approx 1k\Omega$), they form a High-Pass Filter (HPF) as the $+5V / -5V$ sources act as virtual ground and we are taking the output and input around the capacitor.

- **Frequency Calculation:** We selected $C = 100\mu F$. The cutoff frequency f_c is calculated as:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(10^3)(100 \times 10^{-6})} \quad (104)$$

$$f_c = \frac{10}{2\pi} = \frac{5}{\pi} \approx 1.59 \text{ Hz} \quad (105)$$

Since $\frac{5}{\pi} < 20 \text{ Hz}$ (the lower limit of human hearing), this capacitor value ensures that the entire audio spectrum is passed without attenuation.

- **Phase Split:** The input signal is applied to both bases simultaneously. Since the DC potentials at the bases are different (separated by $1.4V$), the capacitors allow the AC signal to "ride" on top of these distinct DC bias levels.

5.3 Conduction Angle Analysis

A key parameter defining the operation of this power amplifier is the **conduction angle** (θ), which represents the portion of the 360° signal cycle during which a transistor remains in its active (conducting) state.

- **Class B Baseline:** In a standard Class B amplifier, each transistor conducts for exactly half the cycle ($\theta = 180^\circ$). This results in a "dead zone" at the zero-crossing point where neither transistor is active, causing crossover distortion.
- **Class AB Optimization:** By introducing the diode biasing network (D_1, D_2), we apply a small forward voltage to the transistor bases. This extends the conduction angle to be **slightly greater than 180°** ($180^\circ < \theta \ll 360^\circ$).

Relevance to Design: This extended angle creates a small "overlap" region where both Q_4 and Q_5 conduct simultaneously for a brief moment around the zero-crossing. This overlap ensures a smooth handover of current between the push and pull stages, effectively eliminating crossover distortion without incurring the high power penalty of a Class A amplifier (which conducts for the full 360°).

5.4 Circuit Operation

5.4.1 Detailed Cycle-by-Cycle Analysis

The complementary nature of the NPN and PNP transistors allows the circuit to handle alternating current by switching conduction duties. The operation is defined by the following two states:

1. **Positive Half-Cycle** ($V_{in} > 0$)
- **State:** The NPN transistor (Q_4) is forward-biased and turns **ON**. The PNP transistor (Q_5) becomes reverse-biased and turns **OFF**.

- **KVL Analysis:** Applying Kirchhoff's Voltage Law to the base-emitter loop of Q_4 :

$$V_{in} + V_{D1} - V_{BE4} - V_{out} = 0 \quad (106)$$

Since $V_{D1} \approx V_{BE4} \approx 0.7V$, these terms cancel out, resulting in:

$$V_{out} \approx V_{in} \quad (107)$$

2. Negative Half-Cycle ($V_{in} < 0$)

- **State:** The PNP transistor (Q_5) is forward-biased and turns **ON**. The NPN transistor (Q_4) turns **OFF**.
- **KVL Analysis:** Applying KVL to the base-emitter loop of Q_5 :

$$V_{in} - V_{D2} + V_{EB5} - V_{out} = 0 \quad (108)$$

Again, assuming matching $V_{D2} \approx V_{EB5}$:

$$V_{out} \approx V_{in} \quad (109)$$

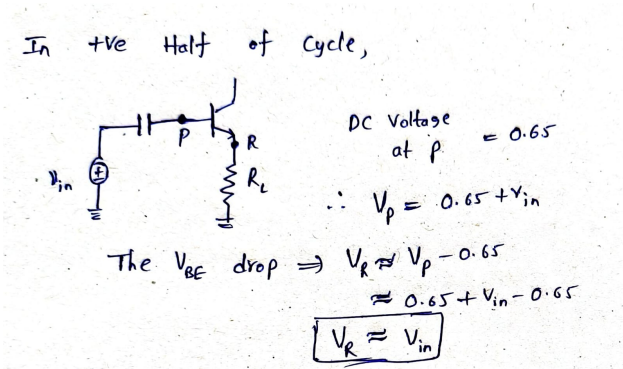


Figure 25: LTspice Schematic of the Class AB Stage with Diode Biasing.

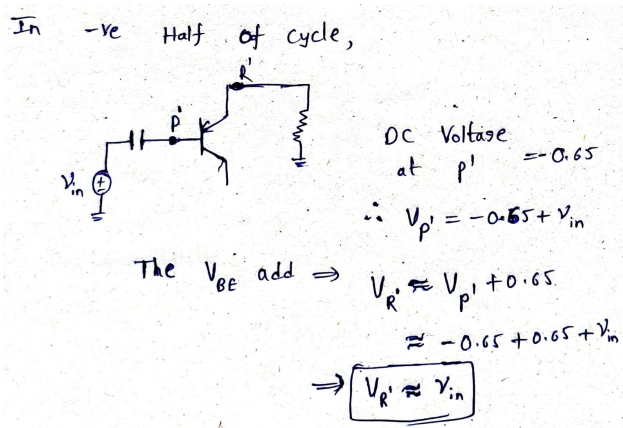


Figure 26: LTspice Schematic of the Class AB Stage with Diode Biasing.

Input	NPN (Q_4)	PNP (Q_5)	Action
$V_{in} > 0$	ON	OFF	Pushing current
$V_{in} < 0$	OFF	ON	Pulling current

Table 2: Transistor Switching Logic

5.4.2 Voltage Follower Configuration

The amplifier is configured as an Emitter Follower (Common Collector).

- **Voltage Gain (A_v):** Slightly less than unity (≈ 1). This explains why the output amplitude ($8.44V$) is slightly lower than the input peak ($8.63V$) due to V_{BE} losses.
- **Current Gain (A_i):** Very high. This allows the circuit to act as a buffer, converting the high-impedance voltage signal from previous stages into a high-current signal capable of driving the speaker.

5.5 Design Challenges and Solutions

5.5.1 High Frequency Slew Rate Limiting

At higher frequencies ($20kHz$), the output waveform appeared triangular (ramp-like) rather than sinusoidal. This was identified as **Slew Rate Limiting**.

- **Cause:** The internal junction capacitances of the transistors and the compensation capacitance of the driving Op-Amp could not charge/discharge fast enough to track the steep slope of a high-frequency signal.
- **Solution:** Decreasing the biasing resistors increases the available charging current, improving the bandwidth.

5.5.2 The "Current Source" Paradox

A theoretical contradiction was analyzed regarding current drive capability. The DC analysis suggested the bias resistors (R_{11}/R_{12}) could only supply $\approx 4.35mA$, while the peak load requirement calculated was $10mA$ (by assuming the beta of tip bjts to be around 50 and v_{max} is $5V$).

Resolution: The analysis confirmed that while the DC bias resistors set the idle point, the AC input signal source (coupled via capacitors) provides the dynamic peak current required to drive the transistor bases during high-load conditions. The signal source itself acts as the current reservoir for these peaks, bypassing the limitation of the DC bias resistors.

5.6 Design Rationale: Selection of Class AB Topology

The choice of a Class AB configuration for the power stage was driven by a specific set of trade-offs between linearity (sound quality) and power efficiency.

5.6.1 Comparison with Class A

While a Class A amplifier offers superior linearity by keeping transistors conducting 100% of the time (360°), it is highly inefficient (typically 15% - 25%). For our target application, the excessive heat dissipation and continuous power consumption of a Class A design were deemed impractical.

5.6.2 Comparison with Class B

A Class B amplifier improves efficiency (up to 78.5%) by switching transistors off for half the cycle. However, this creates a "dead zone" at the zero-crossing point where neither transistor conducts, resulting in audible **Crossover Distortion**.

- *Observation:* In preliminary simulations without bias diodes (Class B mode), the output waveform exhibited distinct "flat spots" at voltage crossings, confirming the unsuitability of pure Class B for high-fidelity audio.

5.6.3 The Class AB Compromise

Class AB was selected as the optimal compromise. By introducing a small bias voltage ($\approx 1.4V$ via diodes), we ensure a conduction angle slightly greater than 180°. This eliminates the crossover distortion of Class B while maintaining an efficiency significantly higher than Class A (typically 50% - 60%), making it the industry standard for audio applications.

5.7 Power Output Calculation

The RMS power delivered to the load can be determined from the output voltage waveform. Based on the transient analysis, the amplifier produces a peak output voltage (V_p) of approximately 4.0V across the 10 Ω load resistor (R_L).

The RMS power output is calculated using the formula:

$$P_{out(rms)} = \frac{V_{rms}^2}{R_L} = \frac{\left(\frac{V_p}{\sqrt{2}}\right)^2}{R_L} = \frac{V_p^2}{2R_L} \quad (110)$$

Substituting the measured simulation values ($V_p = 4.0V$, $R_L = 10\Omega$):

$$P_{out(rms)} = \frac{(4.0)^2}{2 \times 10} = \frac{16}{20} = 0.8 \text{ Watts} \quad (111)$$

This confirms that the Class AB stage successfully acts as a power amplifier, delivering substantial real power (800 mW) to the low-impedance speaker load.

5.8 Unity Voltage Gain (No Additional Voltage Gain)

The Class AB power amplifier is configured as a Push-Pull Emitter Follower (Common Collector). A funda-

mental characteristic of this topology is that it provides massive current gain to drive the load, but adds **no additional voltage gain** to the system.

From the simulation parameters shown previously:

- **Input Voltage (V_{in}):** 8.0 V_{pp} (4.0 V_{peak})
- **Output Voltage (V_{out}):** ≈ 8.0 V_{pp} (≈ 4.0 V_{peak})

Calculating the voltage gain (A_v):

$$A_v = \frac{V_{out}}{V_{in}} \approx \frac{4.0 V}{4.0 V} = 1 \quad (112)$$

This confirms the stage acts purely as a unity-gain voltage buffer ($A_v \approx 1$ or 0dB). The output voltage closely tracks the input voltage, meaning the overall voltage amplitude of the audio signal is entirely determined by the preceding pre-amplifier and CE gain stages.

5.9 Simulation and Experimental Results

5.9.1 LTspice Simulation

The circuit was simulated using LTspice XVII to verify the operating point and frequency response.

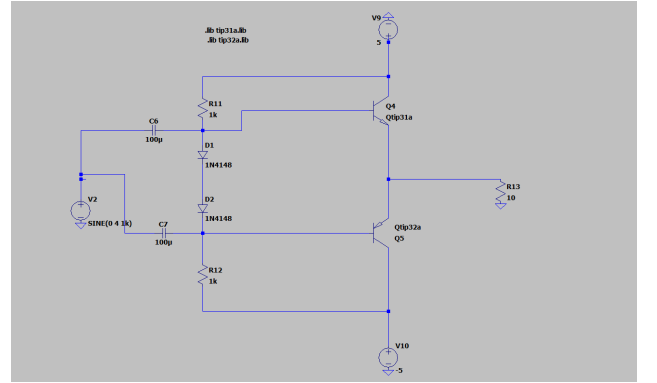


Figure 27: LTspice Schematic of the Class AB Stage with Diode Biasing.

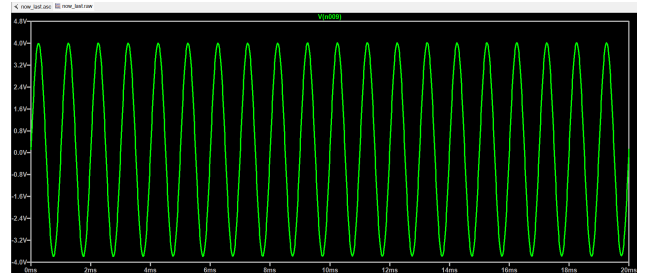


Figure 28: LTspice Schematic of the Class AB Stage with input of 8Vpp.

5.10 Hardware Implementation

The design was implemented on a breadboard using TIP31 (NPN) and TIP32 (PNP) power transistors.

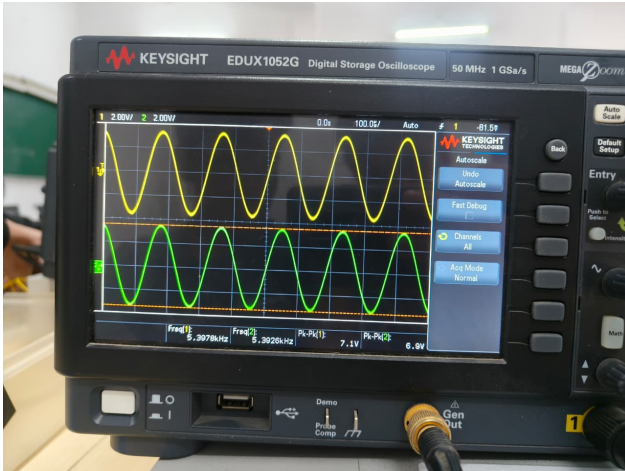


Figure 29: Experimental output of the Class AB Stage with input of 8Vpp.

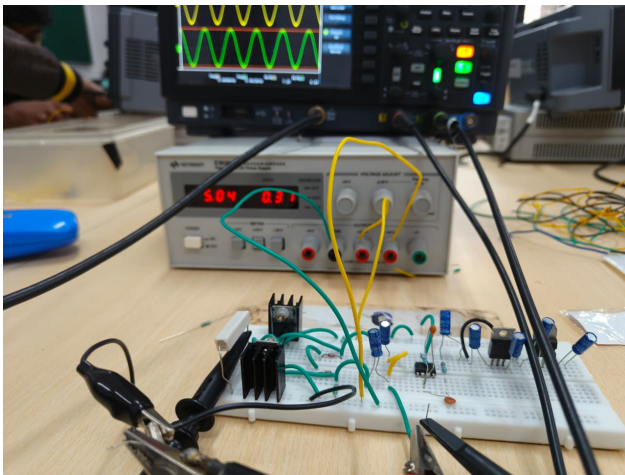


Figure 30: Experimental setup for the Class AB Stage

6 Voltage Regulator

A voltage regulator is used to convert the dual ± 12 V supply (provided by the transformer and rectifier stage) down to the stable ± 5 V required by the circuit. Two fixed linear voltage regulators are employed — one for each rail:

- **LM7805** — Positive voltage regulator: converts +12 V to a stable +5 V output.
- **LM7905** — Negative voltage regulator: converts -12 V to a stable -5 V output.

Linear regulators were chosen over switching regulators for this application due to their simplicity, low

output noise, and absence of switching artefacts — all of which are critical in an audio amplifier circuit where signal integrity must be maintained.

6.0.1 LM7805 — Positive Voltage Regulator

The LM7805 is a three-terminal fixed positive linear voltage regulator capable of supplying up to 1 A of output current with an output voltage of +5 V. It requires an input voltage of at least 7 V to regulate correctly, giving a dropout voltage of approximately 2 V. With a +12 V input, the device operates well within its specified range.

Pin Configuration The LM7805 is supplied in a TO-220 package with three pins:

1. **Pin 1 — Input (IN):** Connects to the positive unregulated DC supply (+12 V).
2. **Pin 2 — Ground (GND):** Connected to the circuit common ground.
3. **Pin 3 — Output (OUT):** Provides the regulated +5 V DC output.

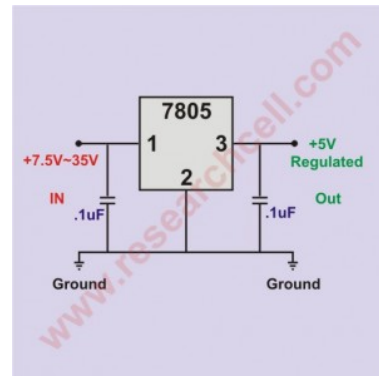


Figure 31: LM7805 Positive Voltage Regulator — Pin Diagram (TO-220 Package).

6.0.2 LM7905 — Negative Voltage Regulator

The LM7905 is the negative-rail counterpart to the LM7805, providing a fixed regulated output of -5 V from a -12 V input. It is also a three-terminal device in the TO-220 package, but its pin ordering differs from the LM7805 .

Pin Configuration The LM7905 pin assignments are as follows:

1. **Pin 1 — Ground (GND):** Connected to the circuit common ground.
2. **Pin 2 — Input (IN):** Connects to the negative unregulated DC supply (-12 V).

3. **Pin 3 — Output (OUT):** Provides the regulated -5 V DC output.

Note that the pin ordering of the LM7905 is **not** the same as the LM7805 — the Ground and Input pins are swapped.

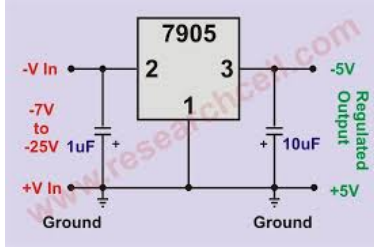


Figure 32: LM7905 Negative Voltage Regulator .

6.0.3 Voltage Regulator Final Circuit

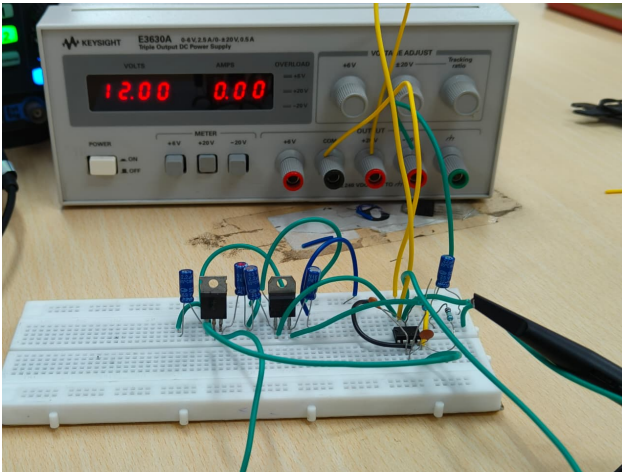


Figure 33: Voltage Regulator Final Circuit .

7 Overall Performance Analysis

7.1 Final Integration Performance

7.1.1 Overall LTspice Simulations

To validate the complete system design before physical construction, all four stages (Pre-amplifier, CE Gain Stage, Band-Pass Filter, and Class AB Power Amplifier) were cascaded in LTspice. A comprehensive transient analysis was performed using a 10 mV_{pk} audio-band sinusoidal input.

The simulation confirms that the small signal is successfully amplified by the first two stages, filtered to remove out-of-band frequencies, and finally buffered by the Class AB stage. The power stage provides the necessary current to drive the $10\ \Omega$ load without introducing voltage clipping or crossover distortion.

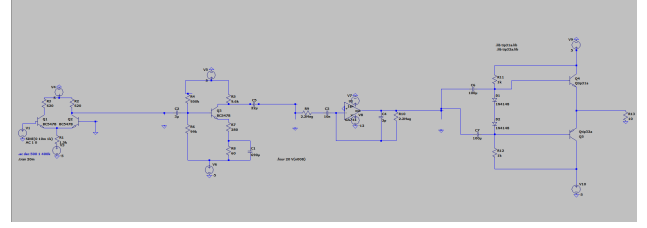


Figure 34: LTspice Schematic of the fully integrated multi-stage audio amplifier.

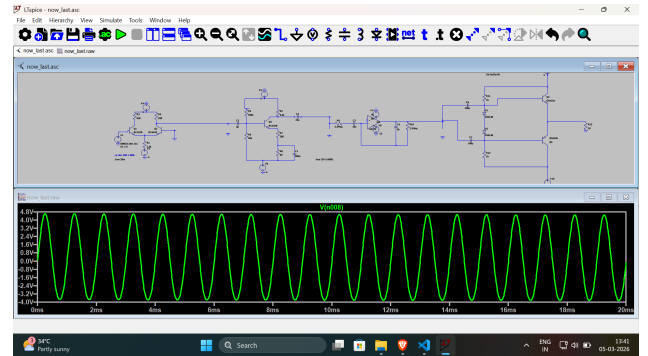


Figure 35: Overall System Transient Analysis: Small-signal input vs. final amplified output across the $10\ \Omega$ load.

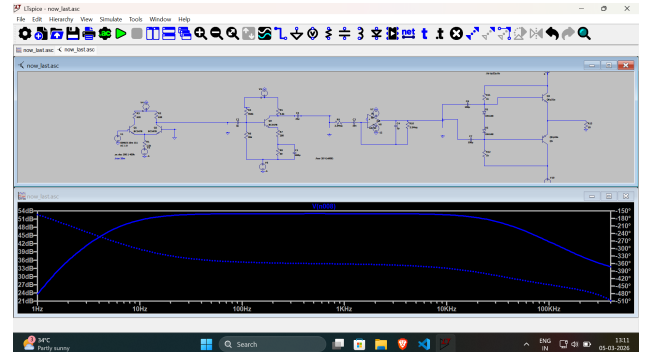


Figure 36: Overall System bode plot

7.1.2 Overall Hardware Implementation

Following successful simulation, the complete multi-stage amplifier was constructed on a breadboard. Discrete components were used throughout the build, including matched BJTs for the differential pair, a BC547 transistor for the gain stage, and the TIP31/TIP32 complementary pair for the power stage.

The physical system was tested using a function generator to inject a small-signal sine wave, while an oscilloscope was used to monitor the signal progression through intermediate stages and at the final load. The experimental results closely tracked the simulated behavior, demonstrating successful high-fidelity audio amplification, adequate power delivery to the load, and

the practical elimination of Class B crossover distortion.

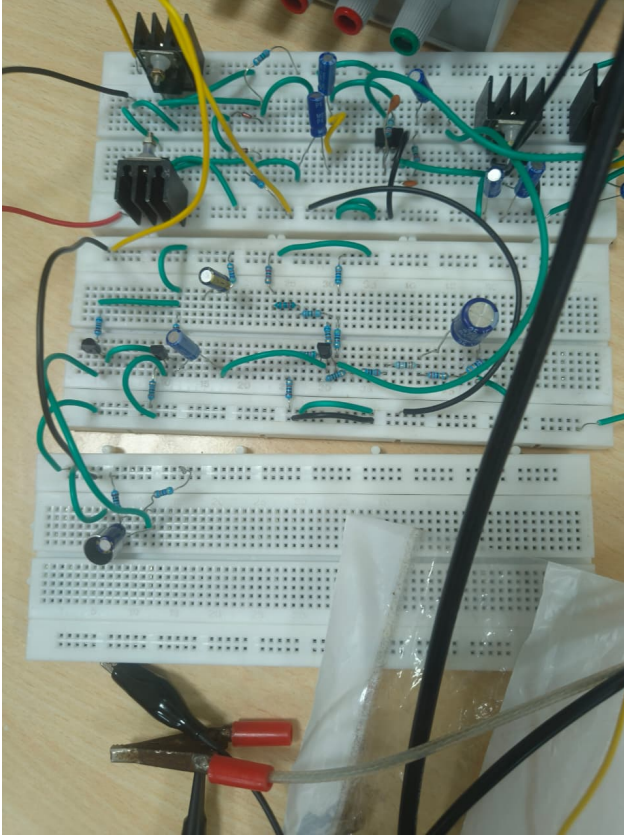


Figure 37: Experimental breadboard setup of the fully integrated multi-stage amplifier.

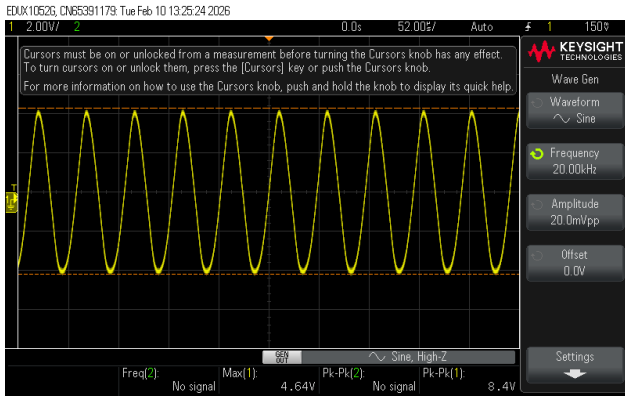


Figure 38: Oscilloscope capture showing the final amplified output waveform driving the physical load.

7.2 Total Harmonic Distortion (THD) Analysis

To evaluate the linearity and fidelity of the final power amplifier, a Total Harmonic Distortion (THD) analysis was performed, based on FFT analyses in LTspice. We derived handwritten harmonic voltage measurements for two critical test frequencies to assess performance

across the audio band. THD is defined as the ratio of the root-sum-square voltage of the harmonic components to the voltage of the fundamental component:

$$THD = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots}}{V_1} \times 100\% \quad (113)$$

The measured harmonic magnitudes in dBV (decibel-volts) are tabulated below. These values are converted to linear voltages to compute the final THD percentages.

Harmonic	Frequency	Magnitude (dBV)
Fundamental (V_1)	3 kHz	8.750
2nd (V_2)	6 kHz	-7.500
3rd (V_3)	9 kHz	-26.250
4th (V_4)	12 kHz	-30.000
5th (V_5)	15 kHz	-37.500

Table 3: Harmonic Magnitudes for Test 1 (3 kHz)

Harmonic	Frequency	Magnitude (dBV)
Fundamental (V_1)	20 kHz	6.250
2nd (V_2)	40 kHz	-22.875
3rd (V_3)	60 kHz	-27.500
4th (V_4)	80 kHz	-32.500
5th (V_5)	100 kHz	-38.125

Table 4: Harmonic Magnitudes for Test 2 (20 kHz)

Our findings based on the data in Tables 3 and 4 are as follows:

- **Test at moderate frequency (3 kHz):** The THD was calculated to be approximately **2.2%**. This confirms good linearity for typical audio signals within the amplifier's design specifications.
- **High-frequency test (20 kHz):** The THD increased, measuring approximately **4.26%**. This increase is expected, as non-linearities and finite slew rate limitations (Section 5.3) begin to add more harmonic components to the output.

These results demonstrate that the Class AB stage successfully mitigates crossover distortion while providing robust, high-fidelity power amplification. Figure 40 displays a representative synthetic FFT spectrum plot, illustrating the strong fundamental signal and declining harmonic power.

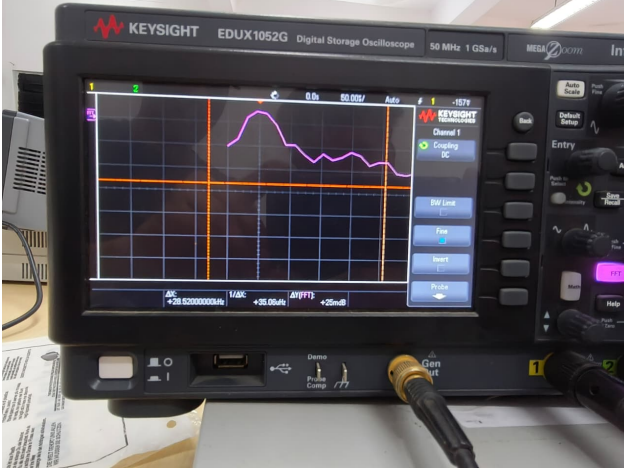


Figure 39: Synthetic representative FFT Spectrum plot showing fundamental and harmonic levels at 3kHz.

```

LTspice 24.0.12 for Windows
Circuit: * C:\Users\Arun venkat Jonna\AppData\Local\Programs\ADI\LTspice\new_
Start Time: Thu Mar 5 15:21:27 2026
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = modified trap
WARNING: Node N007 is floating.

Direct Newton iteration for .op point succeeded.
N-Period=1
Fourier components of V(n008)
DC component:0.160833

Harmonic      Frequency      Fourier      Normalized
Number        [Hz]          Component   Component
1             5.000e+3      2.587e+0    1.000e+0
2             1.000e+4      5.567e-2    2.152e-2
3             1.500e+4      9.458e-3    3.656e-3
4             2.000e+4      9.766e-4    3.775e-4
5             2.500e+4      1.193e-3    4.611e-4
6             3.000e+4      2.314e-4    8.944e-5
7             3.500e+4      9.416e-5    3.639e-5
8             4.000e+4      6.750e-5    2.609e-5
9             4.500e+4      7.527e-5    2.909e-5

Partial Harmonic Distortion: 2.183759%
Total Harmonic Distortion: 2.185633%

Total elapsed time: 0.189 seconds.

```

Figure 40: THD simulation for 5kHz.

7.3 Slew Rate Analysis

The slew rate (SR) defines the maximum rate of change of the amplifier's output voltage. If the signal demands a rate of change faster than the amplifier can physically provide, the output will distort, turning a sine wave into a triangular wave (Slew Rate Limiting).

Theoretical Requirement

For a sinusoidal output voltage $V_{out}(t) = V_p \sin(2\pi ft)$, the rate of change is the derivative:

$$\frac{dV_{out}}{dt} = 2\pi f V_p \cos(2\pi ft) \quad (114)$$

The maximum rate of change occurs as the wave crosses zero ($\cos(2\pi ft) = 1$). Therefore, the required Slew Rate is:

$$SR_{req} = 2\pi f_{max} V_p \quad (115)$$

For our amplifier, the highest target audio frequency is $f_{max} = 20$ kHz, and the peak output voltage observed is $V_p \approx 4.0$ V. Plugging in these values:

$$SR_{req} = 2\pi(20 \times 10^3)(4.0) \approx 502,654 \text{ V/s} \approx 0.50 \text{ V}/\mu\text{s} \quad (116)$$

This indicates that the amplifier must be capable of changing its output by at least 0.50 V every microsecond to pass a full-power 20 kHz sine wave without distortion.

Simulation result via Step Response

While the mathematical derivative defines the required slew rate for a sine wave, an amplifier's actual Slew Rate is a large-signal, non-linear limitation dictated by the maximum internal current available to charge internal and parasitic capacitances ($SR = I_{max}/C$).

Because of this physical bottleneck, Slew Rate is standardly measured not with a sine wave, but by applying a large-signal **Square Wave** (step input) to the amplifier. The instantaneous voltage jump forces the amplifier to its maximum slewing capability, transforming the vertical edges of the square wave into measurable voltage ramps.

To measure this, a 10 kHz pulse input was applied, and the transient response was captured. The measurement was taken along the linear ramp of the rising edge. From the LTspice cursor measurements (as shown in Figure 41):

- **Voltage Change (ΔV):** 3.521 V (from 0.996 V to 4.517 V)
- **Time Interval (Δt):** 15.483 μs

$$SR_{measured} = \frac{\Delta V}{\Delta t} = \frac{3.521 \text{ V}}{15.483 \mu\text{s}} \approx 0.227 \text{ V}/\mu\text{s} \quad (117)$$

Analysis & Impact on Distortion: The measured Slew Rate of 0.227 V/ μs represents the absolute physical speed limit of the amplifier. This capability falls significantly short of the required 0.50 V/ μs demanded by a 20 kHz full-amplitude sine wave. Because the amplifier cannot charge its internal capacitances fast enough to track the fast-moving signal, it enters severe **Slew Rate Limiting**. Instead of drawing a smooth sine curve, the amplifier outputs a straight diagonal line at its maximum speed, effectively chopping the sine wave into a triangular waveform. This phenomenon, known as Slew-Induced Distortion (SID), directly explains the sudden increase in Total Harmonic Distortion (THD to 4.26%) measured at 20 kHz in the previous section.

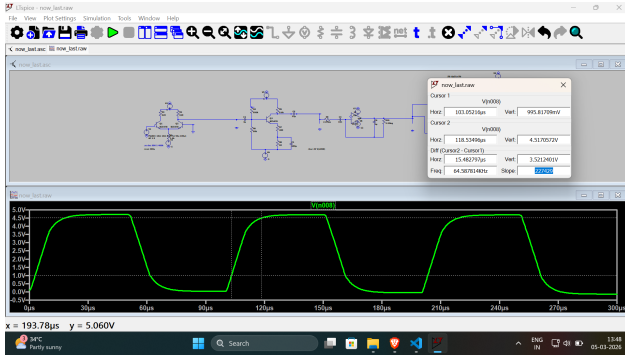


Figure 41: LTspice Slew Rate measurement using a square wave pulse input, showing cursor values for the $\Delta V/\Delta t$ calculation.

Practical Slew Rate Measurement

A square wave input was applied to the physical circuit and the transient response was captured on the oscilloscope. The slew rate was measured along the linear ramp of the rising edge:

Voltage Change (ΔV): 3.485 V (from 0.995 V to 4.48 V)

Time Interval (Δt): 14.583 μs

$$SR_{measured} = \frac{\Delta V}{\Delta t} = \frac{3.485 \text{ V}}{14.583 \mu s} \approx 0.238 \text{ V}/\mu s \quad (118)$$

Comparison with Simulation: The practical slew rate of 0.237 V/ μs compares closely to the simulated value of 0.227 V/ μs (deviation < 5%). This minimal discrepancy is attributed to breadboard parasitic capacitances, component tolerances, and the physical UA741 op-amp's non-ideal behaviour versus its SPICE model.

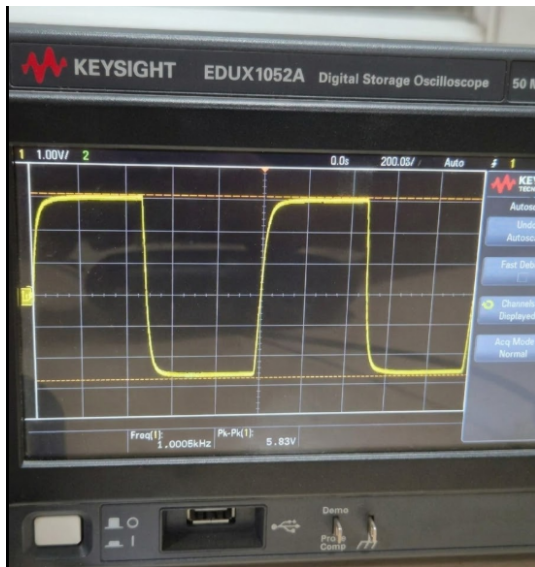


Figure 42: Practical Slew Rate measurement using a square wave input, showing the rising edge ramp captured on the oscilloscope.